GAUSS - LAPLACE



## Tabelul A

Valorile funcției $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$.

| x | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0,0 | 0,5000 | 0,5040 | 0,5080 | 0,5120 | 0,5160 | 0,5199 | 0,5239 | 0,5279 | 0,5319 | 0,5359 |
| 0,1 | 5398 | 5438 | 5478 | 5517 | 5557 | 5596 | 5636 | 5675 | 5714 | 5753 |
| 0,2 | 5793 | 5832 | 5871 | 5910 | 5948 | 5986 | 6026 | 6064 | 6103 | 6141 |
| 0,3 | 6179 | 6217 | 6255 | 6293 | 6331 | 5987 | 6406 | 6443 | 6480 | 6517 |
| 0,4 | 6554 | 6591 | 6628 | 6664 | 6700 | 6736 | 6772 | 6808 | 6844 | 6879 |
| 0,5 | 6915 | 6950 | 6985 | 7019 | 7054 | 7088 | 7123 | 7157 | 7109 | 7224 |
| 0,6 | 7257 | 7291 | 7324 | 7357 | 7389 | 7422 | 7454 | 7486 | 7517 | 7549 |
| 0,7 | 7580 | 7611 | 7642 | 7673 | 7703 | 7734 | 7764 | 7794 | 7823 | 7852 |
| 0,8 | 7881 | 7910 | 7939 | 7967 | 7995 | 8023 | 8051 | 8078 | 8106 | 8133 |
| 0,9 | 8159 | 8186 | 8212 | 8238 | 8264 | 8289 | 8315 | 8340 | 8365 | 8389 |
| 1,0 | 8413 | 8438 | 8461 | 8485 | 8508 | 8531 | 8554 | 8577 | 8599 | 8621 |
| 1,1 | 8643 | 8665 | 8686 | 8708 | 8729 | 8749 | 8770 | 8770 | 8790 | 8810 |
| 1,2 | 8849 | 8869 | 8888 | 8907 | 8925 | 8944 | 8962 | 8980 | 8997 | 9015 |
| 1,3 | 9032 | 9049 | 9066 | 9082 | 9099 | 9115 | 9131 | 9147 | 9162 | 9177 |
| 1,4 | 9192 | 9207 | 9222 | 9236 | 9251 | 9265 | 9279 | 9292 | 9306 | 9319 |
| 1,5 | 9332 | 9345 | 9357 | 9370 | 9382 | 9394 | 9406 | 9418 | 9429 | 9441 |
| 1,6 | 9452 | 9463 | 9474 | 9484 | 9495 | 9505 | 9515 | 9525 | 9535 | 9545 |
| 1,7 | 9554 | 9564 | 9573 | 9582 | 9591 | 9599 | 9608 | 9616 | 9625 | 9633 |
| 1,8 | 9641 | 9649 | 9654 | 9664 | 9671 | 9678 | 9686 | 9693 | 9699 | 9706 |
| 1,9 | 9713 | 9719 | 9726 | 9732 | 9738 | 9744 | 9750 | 9756 | 9761 | 9767 |
| 2,0 | 9772 | 9778 | 9783 | 9788 | 9793 | 9798 | 9803 | 9808 | 9812 | 9817 |
| 2,1 | 9821 | 9826 | 9830 | 9834 | 9838 | 9842 | 9846 | 9850 | 9854 | 9857 |
| 2,2 | 9861 | 9864 | 9868 | 9871 | 9875 | 9878 | 9881 | 9884 | 9887 | 9890 |
| 2,3 | 9893 | 9896 | 9898 | 9901 | 9904 | 9906 | 9909 | 9911 | 9913 | 9916 |
| 2,4 | 9918 | 9920 | 9922 | 9925 | 9927 | 9929 | 9931 | 9932 | 9934 | 9936 |
| 2,5 | 9938 | 9940 | 9941 | 9943 | 9945 | 9946 | 9948 | 9949 | 9951 | 9952 |
| 2,6 | 9953 | 9955 | 9956 | 9957 | 9959 | 9960 | 9961 | 9962 | 9963 | 9964 |
| 2,7 | 9965 | 9966 | 9967 | 9968 | 9969 | 9970 | 9971 | 9972 | 9973 | 9974 |
| 2,8 | 9974 | 9975 | 9976 | 9977 | 9977 | 9978 | 9979 | 9979 | 9980 | 9981 |
| 2,9 | 9981 | 9982 | 9982 | 9983 | 9984 | 9984 | 9985 | 9985 | 9986 | 9986 |
| 3,0 | 9987 | 9990 | 9993 | 9995 | 9997 | 9998 | 9998 | 9999 | 9999 | 1,0000 |

**Cuantilele inferioare $t_{n,p}$ ale repartiției Student: $P(t < t_{n,p}) = p$**

| $n \backslash p$ | 0,7 | 0,75 | 0,8 | 0,9 | 0,95 | 0,975 | 0,990 | 0,995 | 0,999 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0,7265 | 1,0000 | 1,3764 | 3,077 | 6,313 | 12,70 | 31,82 | 63,65 | 31,82 |
| 2 | 0,6162 | 0,8165 | 1,0607 | 1,885 | 2,920 | 4,302 | 6,964 | 9,925 | 6,964 |
| 3 | 0,5844 | 0,7649 | 0,9785 | 1,637 | 2,353 | 3,182 | 4,540 | 5,840 | 4,540 |
| 4 | 0,5686 | 0,7407 | 0,9410 | 1,533 | 2,131 | 2,776 | 3,746 | 4,604 | 3,746 |
| 5 | 0,5594 | 0,7267 | 0,9195 | 1,475 | 2,015 | 2,570 | 3,364 | 4,032 | 3,364 |
| 6 | 0,5534 | 0,7111 | 0,9057 | 1,439 | 1,943 | 2,446 | 3,142 | 3,707 | 3,142 |
| 7 | 0,5491 | 0,7111 | 0,8960 | 1,414 | 1,894 | 2,364 | 2,997 | 3,499 | 2,997 |
| 8 | 0,5459 | 0,7064 | 0,8889 | 1,396 | 1,859 | 2,306 | 2,896 | 3,355 | 2,896 |
| 9 | 0,5435 | 0,7027 | 0,8834 | 1,383 | 1,833 | 2,262 | 2,821 | 3,249 | 2,821 |
| 10 | 0,5415 | 0,6998 | 0,8791 | 1,372 | 1,812 | 2,228 | 2,763 | 3,169 | 2,763 |
| 11 | 0,5399 | 0,6974 | 0,8755 | 1,363 | 1,795 | 2,201 | 2,718 | 3,105 | 2,718 |
| 12 | 0,5386 | 0,6955 | 0,8726 | 1,356 | 1,782 | 2,178 | 2,681 | 3,054 | 2,610 |
| 13 | 0,5375 | 0,6938 | 0,8702 | 1,350 | 1,770 | 2,160 | 2,650 | 3,012 | 2,650 |
| 14 | 0,5366 | 0,6924 | 0,8681 | 1,345 | 1,761 | 2,144 | 2,624 | 2,976 | 2,624 |
| 15 | 0,5357 | 0,6912 | 0,8662 | 1,340 | 1,753 | 2,131 | 2,602 | 2,946 | 2,602 |
| 16 | 0,5350 | 0,6901 | 0,8647 | 1,336 | 1,745 | 2,119 | 2,583 | 2,920 | 2,583 |
| 17 | 0,5344 | 0,6892 | 0,8633 | 1,333 | 1,739 | 2,109 | 2,566 | 2,898 | 2,566 |
| 18 | 0,5338 | 0,6884 | 0,8620 | 1,330 | 1,734 | 2,100 | 2,552 | 2,878 | 2,552 |
| 19 | 0,5333 | 0,6876 | 0,8610 | 1,327 | 1,729 | 2,093 | 2,539 | 2,860 | 2,539 |
| 20 | 0,5329 | 0,6870 | 0,8600 | 1,325 | 1,724 | 2,086 | 2,528 | 2,845 | 2,528 |
| 21 | 0,5325 | 0,6864 | 0,8591 | 1,323 | 1,720 | 2,079 | 2,517 | 2,831 | 2,517 |
| 22 | 0,5321 | 0,6858 | 0,8583 | 1,321 | 1,717 | 2,073 | 2,508 | 2,818 | 2,508 |
| 23 | 0,5317 | 0,6853 | 0,8575 | 1,319 | 1,713 | 2,068 | 2,499 | 2,807 | 2,499 |
| 24 | 0,5314 | 0,6848 | 0,8569 | 1,317 | 1,710 | 2,063 | 2,492 | 2,797 | 2,492 |
| 25 | 0,5312 | 0,6844 | 0,8562 | 1,316 | 1,708 | 2,059 | 2,485 | 2,787 | 2,485 |
| 26 | 0,5309 | 0,6840 | 0,8557 | 1,315 | 1,705 | 2,055 | 2,478 | 2,778 | 2,478 |
| 27 | 0,5306 | 0,6837 | 0,8551 | 1,313 | 1,703 | 2,051 | 2,472 | 2,770 | 2,472 |
| 28 | 0,5304 | 0,6834 | 0,8546 | 1,312 | 1,701 | 2,048 | 2,467 | 2,763 | 2,467 |
| 29 | 0,5302 | 0,6830 | 0,8542 | 1,311 | 1,699 | 2,045 | 2,462 | 2,756 | 2,462 |
| 30 | 0,5300 | 0,6828 | 0,8538 | 1,310 | 1,697 | 2,042 | 2,457 | 2,750 | 2,457 |
| 35 | 0,5292 | 0,6816 | 0,8520 | 1,306 | 1,689 | 2,030 | 2,437 | 2,723 | 2,437 |
| 40 | 0,5286 | 0,6807 | 0,8507 | 1,303 | 1,683 | 2,021 | 2,423 | 2,704 | 2,423 |
| 50 | 0,5278 | 0,6794 | 0,8489 | 1,298 | 1,675 | 2,008 | 2,403 | 2,677 | 2,403 |
| 60 | 0,5272 | 0,6786 | 0,8477 | 1,295 | 1,670 | 2,000 | 2,390 | 2,660 | 2,390 |
| 80 | 0,5265 | 0,6776 | 0,8461 | 1,292 | 1,664 | 1,990 | 2,373 | 2,638 | 2,373 |
| 100 | 0,5261 | 0,6770 | 0,8452 | 1,290 | 1,660 | 1,984 | 2,364 | 2,625 | 2,364 |
| 500 | 0,5247 | 0,6750 | 0,8423 | 1,283 | 1,647 | 1,964 | 2,333 | 2,585 | 2,333 |

$\chi^2$ PEARSON

## 4.5 TABELE STATISTICE



**Cuantilele inferioare** $\chi_{\alpha;n}$ **ale repartiţiei** $\chi_{\alpha,n}$ : $P(X < \chi_{\alpha;n}) = \alpha$

| n/α | ... | α | ... |
|---|---|---|---|
| n | ... | $\chi^2_{\alpha;n}$ | |

| n/α | 0,005 | 0,01 | 0,025 | 0,05 | 0,1 | 0,9 | 0,95 | 0,975 | 0,99 | 0,995 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $0,0^52$ | $0,0^32$ | $0,0^21$ | $0,0^24$ | 0,016 | 2,71 | 3,84 | 5,02 | 6,63 | 7,88 |
| 2 | $0,0^22$ | 0,02 | 0,051 | 0,103 | 0,211 | 4,61 | 5,99 | 7,38 | 9,21 | 10,60 |
| 3 | 0,0243 | 0,115 | 0,216 | 0,352 | 0,584 | 6,25 | 7,81 | 9,35 | 11,34 | 12,84 |
| 4 | 0,091 | 0,297 | 0,484 | 0,711 | 1,06 | 7,78 | 9,49 | 11,14 | 13,28 | 14,86 |
| 5 | 0,210 | 0,554 | 0,831 | 1,15 | 1,61 | 9,24 | 11,07 | 12 83 | 15,09 | 16,75 |
| 6 | 0,381 | 0,872 | 1,24 | 1,64 | 2,20 | 10,64 | 12,59 | 14,45 | 16,81 | 18,55 |
| 7 | 0,599 | 1,24 | 1,69 | 2,17 | 2,83 | 12,02 | 14,07 | 16,01 | 18,48 | 20,28 |
| 8 | 0,857 | 1,65 | 2,18 | 2,73 | 3,49 | 13,36 | 15,51 | 17,53 | 20,09 | 21,95 |
| 9 | 1,15 | 2,09 | 2,70 | 3,32 | 4,17 | 14,68 | 16,92 | 19,02 | 21,67 | 23,59 |
| 10 | 1,48 | 2,56 | 3,25 | 3,94 | 4,86 | 15,99 | 18,31 | 20,48 | 23,21 | 25,19 |
| 12 | 2,21 | 3,57 | 4,40 | 5,23 | 6,30 | 18,55 | 21,03 | 23,34 | 26,22 | 28,30 |
| 14 | 3,04 | 4,66 | 5,63 | 6,57 | 7,79 | 21,06 | 23,68 | 26,12 | 29,14 | 31,32 |
| 16 | 3,94 | 5,81 | 6,91 | 7,96 | 9,31 | 23,54 | 26,30 | 28 85 | 32,00 | 34,27 |
| 18 | 4,90 | 7,02 | 8,23 | 9,39 | 10,86 | 25,99 | 28,87 | 31,53 | 34,81 | 37,16 |
| 20 | 5,92 | 8,26 | 9,59 | 10,85 | 12,44 | 28,41 | 31,41 | 34,17 | 37,57 | 40,00 |
| 25 | 8,65 | 11,52 | 13,12 | 14,61 | 16,47 | 34,38 | 37,65 | 40,65 | 44,31 | 46,93 |
| 30 | 11,59 | 14,95 | 16,79 | 18,49 | 20,60 | 40,26 | 43,77 | 46,98 | 50,89 | 53,67 |
| 40 | 17,92 | 22,16 | 24,43 | 26,51 | 29,05 | 51,81 | 55,76 | 59,34 | 63,69 | 66,77 |
| 50 | 24,67 | 29,71 | 32,36 | 34,76 | 37,69 | 63,17 | 67,50 | 71,42 | 76,15 | 79,49 |
| 60 | 31,74 | 37,48 | 40,48 | 43,19 | 46,46 | 74,40 | 79,08 | 83,30 | 88,38 | 91,95 |
| 100 | 61,92 | 70,06 | 74,22 | 77,93 | 82,36 | 118,5 | 124,3 | 129,6 | 135,8 | 140,2 |

# 4.5 Tabele statistice

I (Gauss)

**Tabel pentru valorile funcţiei de repartiţie $N(0,1)$**

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$$

| x | ... | 4 | ... |
|---|---|---|---|
| 1,6 | ... | 0,9495=F(1,64) | ... |

| z | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0,0 | .5000 | .5039 | .5079 | .5119 | .5159 | .5199 | .5239 | .5279 | .5318 | .5359 |
| 0,1 | .5398 | .5418 | .5477 | .5517 | .5556 | .5596 | .5635 | .5674 | .5714 | .5753 |
| 0,2 | .5792 | .5813 | .5870 | .5909 | .5948 | .5987 | .6025 | .6064 | .6102 | .6140 |
| 0,3 | .6179 | .6201 | ,6255 | .6293 | .6330 | .6368 | .6405 | .6443 | .6480 | .6517 |
| 0,4 | .6554 | .6577 | .6627 | .6664 | .6700 | .6736 | .6772 | .6808 | .6843 | .6879 |
| 0,5 | .6914 | .6937 | .6984 | .7019 | .7054 | .7088 | .7122 | .7156 | .7190 | .7224 |
| 0,6 | .7257 | .7280 | .7323 | .7356 | .7389 | .7421 | .7453 | .7485 | .7517 | .7549 |
| 0,7 | .7580 | .7603 | .7642 | .7673 | .7703 | .7733 | .7763 | .7793 | .7823 | .7852 |
| 0,8 | .7881 | .7903 | .7938 | .7967 | .7995 | .8023 | .8051 | .8078 | .8105 | .8132 |
| 0,9 | .8159 | .8180 | .8212 | .8238 | .8263 | .8289 | .8314 | .8339 | .8364 | .8389 |
| 1,0 | .8413 | .8433 | .8461 | .8484 | .8508 | .8531 | .8554 | .8576 | .8599 | .8621 |
| 1,1 | .8643 | .8661 | .8686 | .8707 | .8728 | .8749 | .8769 | .8790 | .8810 | .8829 |
| 1,2 | .8849 | .8866 | .8887 | .8906 | .8925 | .8943 | .8961 | .8979 | .8997 | .9014 |
| 1,3 | .9032 | .9047 | .9065 | .9082 | .9098 | 9114 | .9130 | .9146 | .9162 | .9177 |
| 1,4 | .9192 | .9205 | .9222 | .9236 | .9250 | .9264 | .9278 | .9292 | .9305 | 9318 |
| 1,5 | .9331 | .9343 | .9357 | .9369 | .9382 | .9394 | .9406 | .9417 | .9429 | .9440 |
| 1,6 | .9452 | .9462 | .9473 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9544 |
| 1,7 | .9554 | .9563 | .9572 | .9581 | .9590 | .9599 | .9607 | .9616 | .9624 | .9632 |
| 1,8 | .9640 | .9648 | .9656 | .9663 | .9671 | .9678 | .9685 | .9692 | .9699 | .9706 |
| 1,9 | .9712 | .9719 | .9725 | .9731 | .9738 | .9744 | .9750 | .9755 | .9812 | .9816 |
| 2,1 | .9821 | .9825 | .9829 | .9834 | .9838 | .9842 | .9846 | .9850 | .9853 | .9857 |
| 2,2 | .9861 | .9864 | .9868 | .9871 | .9874 | .9877 | .9880 | .9884 | .9887 | .9889 |
| 2,3 | .9892 | .9895 | .9898 | .9900 | .9903 | .9906 | .9908 | 9911 | .9913 | .9915 |
| 2,4 | .9918 | .9920 | .9922 | .9924 | .9926 | .9928 | .9930 | .9932 | .9934 | .9936 |
| 2,5 | .9937 | .9939 | .9941 | .9943 | .9944 | .9946 | .9947 | .9949 | .9950 | .9952 |
| 3,0 | .9986 | .9986 | .9987 | .9987 | .9988 | .9988 | .9989 | .9990 | .9989 | .9990 |